

```
In [1]: # First-Order Integration
# Find the integral of
# f(x) = x + 2

from sympy import *

x = symbols('x')
f = x + 2

F_x = integrate(f, x)
print(" F(x) =", F_x)

display(Eq(Symbol('Fx'), F_x))
```

$$F(x) = x^{**2}/2 + 2*x$$

$$Fx = \frac{x^2}{2} + 2x$$

```
In [4]: from sympy import *

x = symbols('x')
f = x**4 + x**3 + x**2 + 2

F_x = integrate(f, x)
print(" F(x) =", F_x)

display(Eq(Symbol('Fx'), F_x))
```

$$F(x) = x^{**5}/5 + x^{**4}/4 + x^{**3}/3 + 2*x$$

$$Fx = \frac{x^5}{5} + \frac{x^4}{4} + \frac{x^3}{3} + 2x$$

```
In [3]: from sympy import *

x = symbols('x')
f = x**2 + x**3

F_x = integrate(f, x)
print(" F(x) =", F_x)

display(Eq(Symbol('Fx'), F_x))
```

$$F(x) = x^{**4}/4 + x^{**3}/3$$

$$Fx = \frac{x^4}{4} + \frac{x^3}{3}$$

In [8]: *#Exponential Function*

```
from sympy import *  
  
x = symbols('x')  
f = exp(x)  
  
F_x = integrate(f, x)  
print(" F(x) =", F_x)  
  
display(Eq(Symbol('Fx'), F_x))
```

$$F(x) = \exp(x)$$

$$Fx = e^x$$

In [9]: *from sympy import **

```
x = symbols('x')  
f = exp(2*x)  
  
F_x = integrate(f, x)  
print(" F(x) =", F_x)  
  
display(Eq(Symbol('Fx'), F_x))
```

$$F(x) = \exp(2x)/2$$

$$Fx = \frac{e^{2x}}{2}$$



In [12]: *from sympy import **

```
x = symbols('x')  
f = exp(4*x)  
  
F_x = integrate(f, x)  
print(" F(x) =", F_x)  
  
display(Eq(Symbol('Fx'), F_x))
```

$$F(x) = \exp(4x)/4$$

$$Fx = \frac{e^{4x}}{4}$$



```
In [14]: #Logarithmic Functions
from sympy import *

x = symbols('x')
f = (1/x)

F_x = integrate(f, x)
print(" F(x) =", F_x)

display(Eq(Symbol('Fx'), F_x))
```

$$F(x) = \log(x)$$
$$Fx = \log(x)$$

```
In [15]: #Logarithmic Functions
from sympy import *

x = symbols('x')
f = log(x)

F_x = integrate(f, x)
print(" F(x) =", F_x)

display(Eq(Symbol('Fx'), F_x))
```

$$F(x) = x \log(x) - x$$
$$Fx = x \log(x) - x$$

```
In [16]: #Logarithmic Functions
from sympy import *

x = symbols('x')
f = log(2*x)

F_x = integrate(f, x)
print(" F(x) =", F_x)

display(Eq(Symbol('Fx'), F_x))
```

$$F(x) = x \log(2x) - x$$
$$Fx = x \log(2x) - x$$

In [17]: *#Trigonometric Functions*

```

from sympy import *

x = symbols('x')
f = sin(x)

F_x = integrate(f, x)
print(" F(x) =", F_x)

display(Eq(Symbol('Fx'), F_x))

```

$$F(x) = -\cos(x)$$

$$Fx = -\cos(x)$$

In [18]: *#Trigonometric Functions*

```

from sympy import *

x = symbols('x')
f = cos(x)

F_x = integrate(f, x)
print(" F(x) =", F_x)

display(Eq(Symbol('Fx'), F_x))

```

$$F(x) = \sin(x)$$

$$Fx = \sin(x)$$

In [21]: *#Trigonometric Functions*

```

from sympy import *

x = symbols('x')
f = sec(x)**2

F_x = integrate(f, x)
print(" F(x) =", F_x)

display(Eq(Symbol('Fx'), F_x))

```

$$F(x) = \sin(x)/\cos(x)$$

$$Fx = \frac{\sin(x)}{\cos(x)}$$

In [23]: *#Trigonometric Functions*

```

from sympy import *

x = symbols('x')
f = csc(x)**2

F_x = integrate(f, x)
print(" F(x) =", F_x)

display(Eq(Symbol('Fx'), F_x))

```

$$F(x) = -\cos(x)/\sin(x)$$

$$Fx = -\frac{\cos(x)}{\sin(x)}$$

In [24]: *#Trigonometric Functions*

```

from sympy import *

x = symbols('x')
f = sec(x)*tan(x)

F_x = integrate(f, x)
print(" F(x) =", F_x)

display(Eq(Symbol('Fx'), F_x))

```

$$F(x) = 1/\cos(x)$$

$$Fx = \frac{1}{\cos(x)}$$

In [25]: *#Trigonometric Functions*

```

from sympy import *

x = symbols('x')
f = csc(x)*cot(x)

F_x = integrate(f, x)
print(" F(x) =", F_x)

display(Eq(Symbol('Fx'), F_x))

```

$$F(x) = -1/\sin(x)$$

$$Fx = -\frac{1}{\sin(x)}$$

In [26]: *#Integration By parts*

```

from sympy import *

x = symbols('x')
f = x**2*exp(x)

F_x = integrate(f, x)
print(" F(x) =", F_x)

display(Eq(Symbol('Fx'), F_x))

```

$$F(x) = (x^2 - 2x + 2) \cdot \exp(x)$$

$$Fx = (x^2 - 2x + 2) e^x$$

In [28]: *#Integration By parts*

```

from sympy import *

x = symbols('x')
f = exp(2*x)*exp(x)

F_x = integrate(f, x)
print(" F(x) =", F_x)

display(Eq(Symbol('Fx'), F_x))

```

$$F(x) = \exp(3x)/3$$

$$Fx = \frac{e^{3x}}{3}$$

In [30]: *#Inverse Trigonometric Functions*

```

from sympy import *

x = symbols('x')
f = 1/(1+x**2)

F_x = integrate(f, x)
print(" F(x) =", F_x)

display(Eq(Symbol('Fx'), F_x))

```

$$F(x) = \operatorname{atan}(x)$$

$$Fx = \operatorname{atan}(x)$$

In [31]: *#Inverse Trigonometric Functions*

```
from sympy import *  
  
x = symbols('x')  
f = 1/sqrt(1-x**2)  
  
F_x = integrate(f, x)  
print(" F(x) =", F_x)  
  
display(Eq(Symbol('Fx'), F_x))
```

$$F(x) = \arcsin(x)$$

$$Fx = \arcsin(x)$$

In [32]: *#Hyperbolic Functions*

```
from sympy import *  
  
x = symbols('x')  
f = sinh(x)  
  
F_x = integrate(f, x)  
print(" F(x) =", F_x)  
  
display(Eq(Symbol('Fx'), F_x))
```

$$F(x) = \cosh(x)$$

$$Fx = \cosh(x)$$

In [33]: *#Hyperbolic Functions*

```
from sympy import *  
  
x = symbols('x')  
f = cosh(x)  
  
F_x = integrate(f, x)  
print(" F(x) =", F_x)  
  
display(Eq(Symbol('Fx'), F_x))
```

$$F(x) = \sinh(x)$$

$$Fx = \sinh(x)$$

In [34]: *#Definite Integrals*

```
from sympy import *  
  
x = symbols('x')  
f = x + 2  
  
F_x = integrate(f, (x, 0, 2))  
print(" F(x) =", F_x)  
  
display(Eq(Symbol('Fx'), F_x))
```

$$F(x) = 6$$

$$Fx = 6$$

In [35]: *#Definite Integrals*

```
from sympy import *  
  
x = symbols('x')  
f = x**3 + x**2 + 2  
  
F_x = integrate(f, (x, 0, 2))  
print(" F(x) =", F_x)  
  
display(Eq(Symbol('Fx'), F_x))
```

$$F(x) = 32/3$$

$$Fx = \frac{32}{3}$$

In [36]: *from sympy import **

```
x = symbols('x')  
f = exp(x)  
  
F_x = integrate(f, (x, 0, 2))  
print(" F(x) =", F_x)  
  
display(Eq(Symbol('Fx'), F_x))
```

$$F(x) = -1 + \exp(2)$$

$$Fx = -1 + e^2$$


```
In [37]: from sympy import *

x = symbols('x')
f = exp(x)*exp(2*x)

F_x = integrate(f, (x, 0, 2))
print(" F(x) =", F_x)

display(Eq(Symbol('Fx'), F_x))
```

$$F(x) = -1/3 + \exp(6)/3$$

$$Fx = -\frac{1}{3} + \frac{e^6}{3}$$

```
In [52]: from sympy import *

x = symbols('x')
f = sin(x)

F_x = integrate(f, (x, 0, 2))
print(" F(x) =", F_x)

display(Eq(Symbol('Fx'), F_x))
```

$$F(x) = 1 - \cos(2)$$

$$Fx = 1 - \cos(2)$$

```
In [56]: from sympy import *

x = symbols('x')
f = x+2

F_x = integrate(f, (x, 1, 2))
print(" F(x) =", F_x)

display(Eq(Symbol('Fx'), F_x))
```

$$F(x) = 7/2$$

$$Fx = \frac{7}{2}$$

```
In [1]: from sympy import *

x = symbols('x')
f = (sqrt(x) + 1/sqrt(x))

F_x = integrate(f, x)
print(" F(x) =", F_x)

display(Eq(Symbol('Fx'), F_x))
```

$$F(x) = 2x^{3/2}/3 + 2\sqrt{x}$$

$$Fx = \frac{2x^{\frac{3}{2}}}{3} + 2\sqrt{x}$$

```
In [3]: from sympy import *

x = symbols('x')
f = x*(sqrt(x) + 1)

F_x = integrate(f, x)
print(" F(x) =", F_x)

display(Eq(Symbol('Fx'), F_x))
```

$$F(x) = 2x^{5/2}/5 + x^2/2$$

$$Fx = \frac{2x^{\frac{5}{2}}}{5} + \frac{x^2}{2}$$

```
In [2]: from sympy import *

x = symbols('x')
f = x*(sqrt(x) + 1)

F_x = integrate(f, x)
print(" F(x) =", F_x)

display(Eq(Symbol('Fx'), F_x))
```

$$F(x) = 2x^{5/2}/5 + x^2/2$$

$$Fx = \frac{2x^{\frac{5}{2}}}{5} + \frac{x^2}{2}$$

```
In [4]: from sympy import *

x = symbols('x')
f = (exp(2*x)+exp(x))/exp(x)

F_x = integrate(f, x)
print(" F(x) =", F_x)

display(Eq(Symbol('Fx'), F_x))
```

$$F(x) = x + \exp(x)$$

$$Fx = x + e^x$$

```
In [5]: from sympy import *

x = symbols('x')
f = (1-x**2)/(1+x**2)

F_x = integrate(f, x)
print(" F(x) =", F_x)

display(Eq(Symbol('Fx'), F_x))
```

$$F(x) = -x + 2*\operatorname{atan}(x)$$

$$Fx = -x + 2 \operatorname{atan}(x)$$

```
In [7]: from sympy import *

x, a = symbols('x a')
f = 1/(sqrt(x+a)+sqrt(x))

F_x = integrate(f, x)
print(" F(x) =", F_x)

display(Eq(Symbol('Fx'), F_x))
```

$$F(x) = 2*a/(3*\sqrt{x} + 3*\sqrt{a+x}) + 2*\sqrt{x}*\sqrt{a+x}/(3*\sqrt{x} + 3*\sqrt{a+x}) + 4*x/(3*\sqrt{x} + 3*\sqrt{a+x})$$

$$Fx = \frac{2a}{3\sqrt{x} + 3\sqrt{a+x}} + \frac{2\sqrt{x}\sqrt{a+x}}{3\sqrt{x} + 3\sqrt{a+x}} + \frac{4x}{3\sqrt{x} + 3\sqrt{a+x}}$$

In [8]: `from sympy import *`

```
x, a = symbols('x a')
f = (a-2*x)**2/3
```

```
F_x = integrate(f, x)
print(" F(x) =", F_x)
```

```
display(Eq(Symbol('Fx'), F_x))
```

$$F(x) = a^2x/3 - 2ax^2/3 + 4x^3/9$$

$$Fx = \frac{a^2x}{3} - \frac{2ax^2}{3} + \frac{4x^3}{9}$$



In [13]: `from sympy import *`

```
x, a = symbols('x a')
f = (1+exp(x))**3/exp(x)
```

```
F_x = integrate(f, x)
print(" F(x) =", F_x)
```

```
display(Eq(Symbol('Fx'), F_x))
```

$$F(x) = 3x + \exp(2x)/2 + 3\exp(x) - \exp(-x)$$

$$Fx = 3x + \frac{e^{2x}}{2} + 3e^x - e^{-x}$$



In [14]: `from sympy import *`

```
x, a = symbols('x a')
f = (1+exp(x))**3/exp(x)
```

```
F_x = integrate(f, x)
print(" F(x) =", F_x)
```

```
display(Eq(Symbol('Fx'), F_x))
```

$$F(x) = 3x + \exp(2x)/2 + 3\exp(x) - \exp(-x)$$

$$Fx = 3x + \frac{e^{2x}}{2} + 3e^x - e^{-x}$$



```
In [4]: from sympy import *

x, y = symbols('x y')
f = x*y + x

F_xy = integrate(f, (x, 0, 3), (y, 0, 4))
print(" F(x) =", F_xy)

display(Eq(Symbol('Fx'), F_xy))
```

$$F(x) = 54$$

$$Fx = 54$$

```
In [6]: from sympy import *

x, y = symbols('x y')
f = x**2 + y**2

F_yx = integrate(f, (y, 0, 2), (x, 0, 1))
print(" F(x) =", F_yx)

display(Eq(Symbol('Fx'), F_yx))
```

$$F(x) = 10/3$$

$$Fx = \frac{10}{3}$$

```
In [7]: from sympy import *

x, y, z = symbols('x y z')
f = x + y + z

F_xyz = integrate(f, (x, 0, 2), (y, 0, 3), (z, 0, 3))
print(" F(x) =", F_xyz)

display(Eq(Symbol('Fx'), F_xyz))
```

$$F(x) = 72$$

$$Fx = 72$$

```
In [8]: from sympy import *

x, y, z = symbols('x y z')
f = x**3 + y**3 + z**3

F_xyz = integrate(f, (x, 0, 2), (y, 0, 3), (z, 0, 3))
print(" F(x) =", F_xyz)

display(Eq(Symbol('Fx'), F_xyz))
```

$$F(x) = 279$$

$$Fx = 279$$

